ICT2217 Network Security

Lab Exercise 2.4: Amplification and Reflection Attacks and Defense

Learning Outcomes

Upon completion of this laboratory exercise, you should be able to:

- Conduct and defend against reflection and amplification DDoS using Smurf attack

Required Hardware

- 1 x Rack of Cisco networking devices
- 1 x Box of cables containing
  - USB-to-RJ45 console cables
  - Ethernet cables
- 3 x Laptops

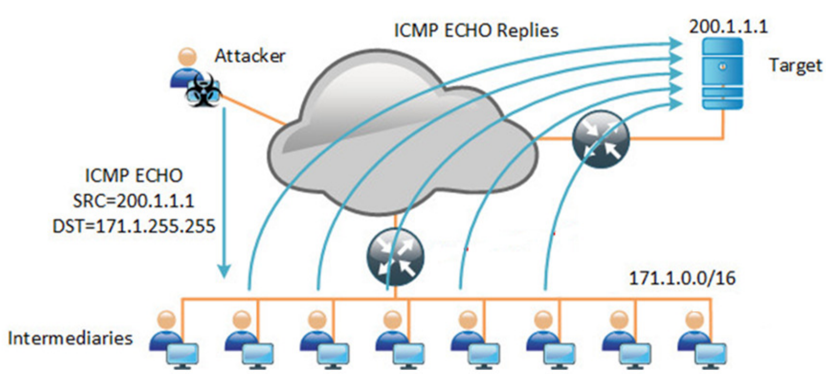
Required Software

- Tera Term https://teratermproject.github.io/index-en.html
- Kali Linux Live Boot on USB drive

PART 1: Conducting Smurf Attack using hping3

1.1 To avoid getting into trouble for misusing NTP or DNS servers in the Internet, we will only experiment with Smurf reflection and amplification attack in this lab exercise.

1.2 Set up the network topology as shown using one router with 3 interfaces, with one interface directly connecting the attacker's PC, another interface directly connecting the target, and the 3rd interface connecting to a switch to interconnect all the intermediaries together.



1.3 To observe the effect of amplification, it is recommended to connect as many intermediary PCs as possible to the switch connected to the router.

1.4 You may boot up the target PC in Windows. The attacker and intermediary PCs must be boot up in **Kali Linux**. If you are running short of Kali Linux USB drives, you may work together with your friends on this lab exercise.

1.5 Configure suitable IP addresses and subnet masks to all the PCs and the router and verify that they are able to ping another. Troubleshoot if necessary.

1.6 Today, Smurf attack is largely defended by default. So, to be able to conduct the attack, you'll need to intentionally disable the default defenses on the router and PCs in Step 1.7 to 1.9.

1.7 At the router, enable the ingress interface to accept IP directed broadcast packets:

```
router(config): interface ingress-interface-id
router(config-if): ip network-broadcast
```

1.8 In addition, configure the egress interface to enable translating IP directed broadcast packets into MAC broadcast frames:

```
router(config): interface egress-interface-id
router(config-if): ip directed-broadcast
```

1.9 At the intermediaries, enable support for broadcast ping as follows:

```
(kali@kali)$ sudo su
(root@kali)# echo 0 > /proc/sys/net/ipv4/icmp_echo_ignore_broadcasts
```

1.10 Start Wireshark at the intermediary and target PCs.

1.11 At the attacker's PC, start a terminal and use hping3 to launch Smurf attack as shown:

```
(root@kali)$ hping3 -1 --flood –a target-IP directed-broadcast-addr
```

For more information on the use of hping3, refer https://www.kali.org/tools/hping3/

1.12 At the intermediaries, can your Wireshark capture the broadcast ping request packets sent by the attacker? If not, verify that your Step 1.7 and 1.8 are configured on the correct interfaces of the router.

1.13 At the target PC, can your Wireshark capture the ping reply packets reflected from the intermediaries? How many times more are there as compared to that sent by the attacker in Step 1.11?

PART 2: Defending against Amplification and Reflection Attacks using ACL

2.1 Configure suitable ACL on the router as discussed.

2.2 When ready, use hping3 to launch Smurf attack again.

2.3 At the intermediaries, can your Wireshark capture the broadcast ping request packets sent by the attacker?

2.4 At the target PC, can your Wireshark capture the ping reply packets reflected from the intermediaries?

2.5 Do you think ACL is effective against amplification and reflection attacks?